# Huawei Access Node MG-ACC-46 Routing Protocol Flap Resolution Procedure

**Node:** MG-ACC-46 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_5115

## Description

The MG-ACC-46 node at the MG Road Core site is experiencing a critical failure due to a routing protocol flap, characterized by BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This failure domain is specific to the Access role on Huawei equipment, where the device is configured as a border router, responsible for establishing and maintaining routing protocol adjacencies with neighboring devices. The underlying protocol mechanics of BGP, OSPF, and ISIS, particularly their interaction with the Huawei device's control plane, contribute to the routing protocol flap. The Huawei device's routing protocol configuration, including its neighbor relationships, timers, and policies, plays a crucial role in this failure domain. Furthermore, the device's hardware resources, such as CPU, memory, and I/O capacity, can also impact the routing protocol's stability and performance. This procedure outlines the steps to diagnose and resolve the routing protocol flap issue on Huawei Access node MG-ACC-46.

## Pre-requisite Checks

- Check: Verify the site access and ensure remote hands are dispatched to MG Road Core, Expected Outcome: Successful site access

- Check: Verify the redundant link status, Expected Outcome: Active redundant link

## Potential Root Causes

| Cause | Probability |
|---|---|
| BGP neighbor relationship issues | High |
| OSPF adjacency loss due to network congestion | Medium |
| ISIS LSP flooding due to hardware resource constraints | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display bgp neighbor, Expected Output: Established neighbor relationships, Description: This command displays the BGP neighbor relationships, including the status of established connections, to help identify potential issues with neighbor relationships.

Command: display ospf peer, Expected Output: Established OSPF peer relationships, Description: This command displays the OSPF peer relationships, including the status of established connections, to help identify potential issues with OSPF adjacencies.

Command: display isis lsp, Expected Output: ISIS LSP information, Description: This command displays the ISIS LSP information, including
```

```
the LSP flooding status, to help identify potential issues with ISIS
LSP flooding.
```

## Resolution Steps

Step 1: Step: Reset BGP neighbor relationships, Command: reset bgp neighbor, Description: This step resets the BGP neighbor relationships, which may help resolve issues with neighbor relationships.

Step 2: Step: Adjust OSPF timers, Command: ospf timer adjust, Description: This step adjusts the OSPF timers, which may help resolve issues with OSPF adjacency loss due to network congestion.

Step 3: Step: Adjust ISIS LSP flooding thresholds, Command: isis lsp flood adjust, Description: This step adjusts the ISIS LSP flooding thresholds, which may help resolve issues with ISIS LSP flooding due to hardware resource constraints.

## Post-Check Verification

```
Command: display bgp neighbor, Expected Output: Established neighbor
relationships, Description: This command verifies that the BGP
neighbor relationships are established and stable.

Command: display ospf peer, Expected Output: Established OSPF peer
relationships, Description: This command verifies that the OSPF peer
relationships are established and stable.

Command: display isis lsp, Expected Output: ISIS LSP information,
Description: This command verifies that the ISIS LSP information is
stable and not flooded.
```

## Roll-back Procedures

- Step: Restore original BGP configuration, Command: config return, Description: This step restores the original BGP configuration, which may help resolve issues with BGP neighbor relationships.

- Step: Restore original OSPF configuration, Command: config return, Description: This step restores the original OSPF configuration, which may help resolve issues with OSPF adjacency loss due to network congestion.

- Step: Restore original ISIS configuration, Command: config return, Description: This step restores the original ISIS configuration, which may help resolve issues with ISIS LSP flooding due to hardware resource constraints.